30/11/23

Lecture # 25

Affine Combination

Affine Combination (AF) n.c

$$y = \text{aff}\{u_1, u_2, \ldots u_n\}$$

$$\iff y = c_1u_1 + c_2u_2 + \ldots c_nu_n$$
$$1 = c_1 + c_2 + \ldots c_n$$

$$\iff \tilde{y} \in \text{span}\{\tilde{u}_1, \tilde{u}_2, \ldots \tilde{u}_n\}$$

$$\tilde{u}_i = \begin{pmatrix} 1 \\ u_i \\ 1 \\ 1 \end{pmatrix}$$ → homogeneous form of a additional 1 at bottom

if $u = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ then $\tilde{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix}$

Affine Independance

Check $\{u_1, \ldots u_n\}$ is A.i ?

$$\iff \{\tilde{u}_1, \ldots \tilde{u}_n\} \text{ is l.i}$$

Q1. Check $\underset{y}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}} \in \text{aff}\left\{\underset{u_1}{\begin{pmatrix} 1 \\ -1 \end{pmatrix}}, \underset{u_2}{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}\right\}$

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$$y \in \text{aff}\{u_1, u_2\} \iff \tilde{y} \in \text{span}\{\tilde{u}_1, \tilde{u}_2\}$$

$$= \left(\begin{array}{cc|c} 1 & 1 & 1 \\ -1 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right)$$

as it is consistent. So $y \in \text{aff}\{u_1, u_2\}$

For $\tilde{y} \in \text{span}\{\tilde{u}_1, \tilde{u}_2\}$, augmented form should result consistent (whether unique or infinite).

Q2. $\left\{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ are a.i?

Sol

Writing form form and augmenting with '0'

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$\therefore$ consistent & every col is pivot. So it is a.i.

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Q. What is affine hull?

Ans. Set of all affine combination a.c of $\{u_1 - u_n\}$ is known as affine hull.

Affine Hull aka Affine Span

Q. Find affine hull of $S = \{u\}$

$$\text{affine}\{S\} = \{c_1 u_1 \mid c_1 = 1\}$$

$$\{u_1\} \longrightarrow \text{point}$$

Q. Find affine hull of $S = \{u_1, u_2\}$

$$\text{affine}\{S\} = \{c_1 u_1 + c_2 u_2 \mid c_1 + c_2 = 1\}$$

$$c_1 = 1 - c_2$$

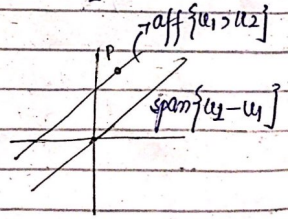
$$= \{u_1(1 - c_2) + c_2 u_2\}$$

$$= u_1 - u_1 c_2 + c_2 u_2$$

$$p \quad u_1 + c_2(u_2 - u_1)$$

$$= p \quad t \quad v$$

$$y = p + tv$$ — line passing through $u_1 \parallel u_2 - u_1$



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Q. Find aff hull of $S = \{u_1, u_2, u_3\}$

$$aff\{S\} = \{c_1u_1 + c_2u_2 + c_3u_3 \mid c_1 + c_2 + c_3 = 1\}$$

$$c_1 = 1 - c_2 - c_3$$

$$= u_1(1 - c_2 - c_3) + c_2u_2 + c_3u_3$$

$$= u_1 - c_2u_1 - c_3u_1 + c_2u_2 + c_3u_3$$

$$= u_1 + c_2(u_2 - u_1) + c_3(u_3 - u_1)$$

is plane passing through u_1 || span$\{u_2 - u_1, u_3 - u_1\}$

Q. Find aff hull of $S = \{u_1, u_2, u_3, u_4\}$

$$aff\{S\} = u_1 + c_2(u_2 - u_1) + c_3(u_3 - u_1) + c_4(u_4 - u_1)$$

is space passing through u_1 || span$\{u_2 - u_1, u_4 - u_1\}$

FLAT:

Translation of a span/subspace is known as flats and each aff(S) is a flat.

dim (flat) = dim of corresponding subspaces.

|| = $(n-1)$

total vectors

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Convex Combination

The convex combination of $\{u_1 -- u_n\}$
is $c_1u_1 + - c_nu_n$
and $c_1 + c_2 + - c_n = 1$ $\Big\}$ conditions
and $c_i \geq 0$

Example:

$$y = \underbrace{2}_{2 \geq 0}u_1 - \underbrace{1}_{-1 \ngeq 0}u_2 \quad = \text{aff comb} \rightarrow 2-1 = 1$$

$$y = 2u_1 + u_2 \quad = \text{l.comb} \rightarrow 2+1 = 3 \neq 1$$

$$y = \tfrac{1}{2}u_1 + \tfrac{1}{2}u_2 \quad = \text{c.comb} \rightarrow \tfrac{1}{2} + \tfrac{1}{2} = 1$$

$$y = 1u_1 + 0u_2 \quad = \text{a.c} + \text{c.c} \rightarrow 1+0 = 1$$

Q. Find convex Hull $\rightarrow$ con(s) if $S = \{u_1\}$

$$\text{con}\{S\} = \{c_1u_1 \mid c_1 = 1, \; c_1 \geq 0\}$$

$$= \{u_1\} \rightarrow \text{point}$$

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→ Find convex Hull of $S = \{u_1, u_2\}$

$\text{Con}\{S\} = \{ c_1u_1 + c_2u_2 \mid c_1 + c_2 = 1$

$(1-c_2)u_1 + c_2u_2$ $\quad c_1 \geq 0$ & $c_2 \geq 0$

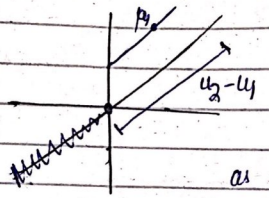
$u_1 - c_2(u_2 - u_1)$

$u_1 - t(u_2 - u_1)$ $\quad c_1 = 1 - c_2$

$0 \leq c_2 \leq 1$

$\downarrow$

t



it results in line segment.

as not an of inf line.

Bilinear Mapping

The mapping $F: U \times V \to W$

$F(u, v)$ is bilinear if

1) $F(c_1u_1 + c_2u_2, v) = c_1F(u_1, v) + c_2F(u_2, v)$

Linear Transformation

$T: \mathbb{R}^n \to \mathbb{R}^n$

- $T(0) = 0$
- $T(u+v) = T(u) + T(v)$
- $T(cv) = cT(v)$

2) $F(u, c_1v_1 + c_2v_2) = c_1F(u, v_1) + c_2F(u, v_2)$

(1) linearity in first component

(2) " " second "

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Bilinear Form:

The bilinear mapping whose codomain is $\boxed{\mathbb{R}}$ is known as bilinear form.

Example:

$$f: \mathbb{R}_2 \times \mathbb{R}_2 \longrightarrow \mathbb{M}_{2\times 2}$$

$$f\left((ax^2+bx+c), \begin{pmatrix} d \\ e \end{pmatrix}\right) = \begin{bmatrix} a+d & b+e \\ c+d & 2e \end{bmatrix}$$

Q. Check whether given mapping is bilinear or not?

1) $f\left(c_1'(a_1x^2+b_1x+c_1) + c_2'(a_2x^2+b_2x+c_2), \begin{pmatrix} d \\ e \end{pmatrix}\right)$

$= f\left((c_1'a_1 + c_2'a_2)x^2 + (c_1'c_1 + c_2'c_2) + (c_1'b_1 + c_2'b_2)x, \begin{pmatrix} d \\ e \end{pmatrix}\right).$

should be 2d for true yes

$$= \begin{bmatrix} c_1'a_1 + c_2'a_2 + d & c_1'b_1 + c_2'b_2 + e \\ & \end{bmatrix}$$

$\therefore$ Answer is ~~False~~ "No! it's not".

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Bilinear Form:

$$f: \mathbb{R}^2 \times \overbrace{\mathbb{R}^3}^{2\times 3 \text{ matrix}} \longrightarrow \mathbb{R}$$

$$f\left(\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} c \\ d \\ e \end{bmatrix}\right) = ac + bd + 2e$$

Let $\{e_1, e_2\}$ are basis of $\mathbb{R}^2$

$\{e_1', e_2', e_3'\}$ " " " $\mathbb{R}^3$

Matrix of bilinear form will be

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} f(e_1, e_1') & f(e_1, e_2') & f(e_1, e_3') \\ f(e_2, e_1') & f(e_2, e_2') & f(e_2, e_3') \end{pmatrix}$$

$$f(e_1, e_1') = f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = 1(1) + 0(0) + 2(0) = 1$$

$$f(e_1, e_2') = f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = 0$$

$$f(e_1, e_3') = f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = 0 + 0 + 2(1) = 2$$

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$$f(e_2, e_1') = f\left(\begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix}\right) = 0$$

$$f(e_2, e_2') = 1$$

$$f(e_2, e_3') = 2$$

$$\begin{bmatrix}1 & 0 & 2\\0 & 1 & 2\end{bmatrix} \rightarrow \text{Matrix of transformation}$$

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